

Emerging Directions in Deep Learning

Examples of ML techniques on particular manifolds (I)

Mihai-Sorin Stupariu, 2025-2026

Introduction

Learning techniques on Lie groups

Applications of symmetric positive definite matrices

Once again: CNN in the 2D-framework

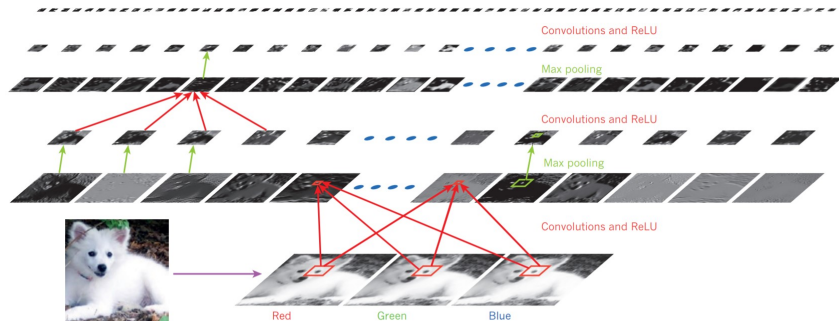


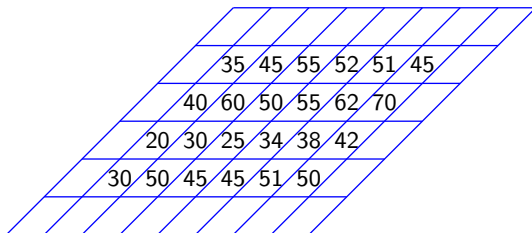
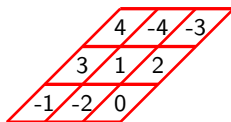
Figure 2 | Inside a convolutional network. The outputs (not the filters) of each layer (horizontally) of a typical convolutional network architecture applied to the image of a Samoyed dog (bottom left; and RGB (red, green, blue) inputs, bottom right). Each rectangular image is a feature map

corresponding to the output for one of the learned features, detected at each of the image positions. Information flows bottom up, with lower-level features acting as oriented edge detectors, and a score is computed for each image class in output. ReLU, rectified linear unit.

[LeCun et al., 2015]

Seek for generalizations...

- ▶ **Aim:** To extend to other categories of geometric objects techniques that work in the 2D-framework.
- ▶ Main operations for a CNN: the grid (regular) structure, the convolution operation).



Recap: several key concepts

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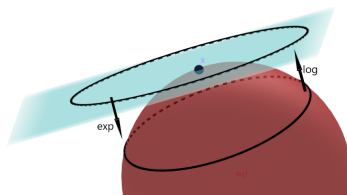
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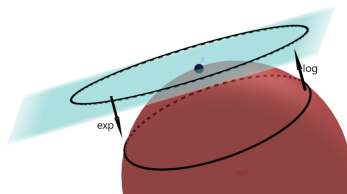
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- ▶ How do we relate the manifold set-up to a ML pipeline (in terms of operations to be performed)?

This lecture...

Two examples of manifolds used in ML pipelines. They have a particular structure, which enables specific computations. Both of them are implemented in Geomstats.

- (1) Lie groups
- (2) Symmetric-positive-definite (SPD) matrices

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- ▶ Fixing an element a of a Lie group G one gets well defined maps $L_a : G \rightarrow G$, $L_a(g) = a \cdot g$ and $R_a : G \rightarrow G$, $R_a(g) = g \cdot a$.

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- ▶ Let e denote the neutral element of G . The tangent space $T_e G$ of G at e naturally has the structure of a **Lie algebra** (usually denoted by $\mathfrak{g}$ - the Lie algebra of G). The Lie algebra $\mathfrak{g}$ is a vector space endowed with an operation called the Lie bracket $[\] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$.

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- ▶ For every $a \in G$, the differentials of the maps L_a, R_a above yield natural maps from $\mathfrak{g}$ to the tangent space $T_a G$.

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- ▶ The special orthogonal group $(SO(n), \cdot)$ - this example is detailed later.

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- ▶ The special orthogonal group $(SO(n), \cdot)$ - this example is detailed later.
- ▶ etc.

The group $(\mathrm{SO}(n), \cdot)$

- By definition

$$\mathrm{SO}(n) = \{A \in \mathcal{M}_n(\mathbb{R}) \mid A \cdot A^T = A^T \cdot A = \mathbb{I}_n, \det(A) = 1\}.$$

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- For $n = 2$ one has the following fact: a matrix 2×2 , say A , is an element of $\mathrm{SO}(2)$ if and only if A has the form

$$A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \quad \text{for a suitable } \alpha.$$

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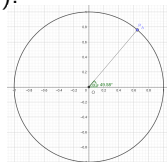
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- Actually A corresponds, in a natural manner, to the rotation of angle α . Thus, the group $\mathrm{SO}(2)$ can be identified, in a natural manner, with the group $(S^1, \cdot)$ (the circle S^1 regarded as the multiplicative group of complex numbers with absolute value 1).



The point $P_A = (\cos \alpha, \sin \alpha) = \cos \alpha + i \sin \alpha$ on the circle S^1 corresponds to the matrix A .

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$$\xi = \begin{pmatrix} 0 & -a \\ a & 0 \end{pmatrix},$$

for a suitable a . Actually giving ξ is equivalent to giving the real number a .

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- Giving an element of $\mathrm{SO}(n)$ is equivalent to giving an orthonormal frame with the same orientation with the canonical one. In turn, this is equivalent to giving an orientation-preserving isometry of the Euclidean space $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_0)$.

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- For $n = 1$ the group reduces to a point, for $n = 2$ one has $\mathrm{SO}(2) \simeq S^1$ (the circle group, the group of 2D rotations). The group $\mathrm{SO}(3)$ is the group of 3D rotations.

A metric on the special orthogonal group

- The Frobenius inner product

$$\langle X_1, X_2 \rangle = \text{tr}(X_1^T X_2), \quad X_1, X_2 \in T_A \text{SO}(n).$$

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- ▶ Try to understand it through examples for $n = 2$ and for $n = 3$!

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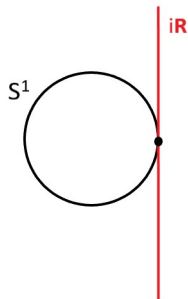
- ▶ One can use the exponential and the logarithm map (take care!) for matrices.
- ▶ Fix $A \in \mathrm{SO}(n)$. One has

$$\exp_A(X) = \exp(XA^T), X \in T_A\mathrm{SO}(n)$$

$$\log_A(B) = \log(AB^T), B \in \mathrm{SO}(n) \text{ "close to } A\text{".}$$

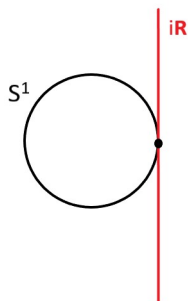
Example: the Lie group S^1 and the exponential map

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- ▶ The exponential map is the usual exponential $i\theta \mapsto e^{i\theta}$.

Application: Lie Group representation for skeletal data

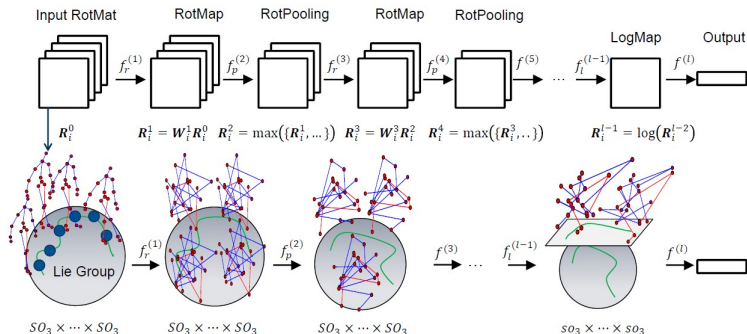


Figure 1. Conceptual illustration of the proposed Lie group Network (LieNet) architecture. In the network structure, the data space of each RotMap/RotPooling layer corresponds to a Lie group, while the weight spaces of the RotMap layers are Lie groups as well.

Source: [Huang et al., 2017](#)

LieTransformer

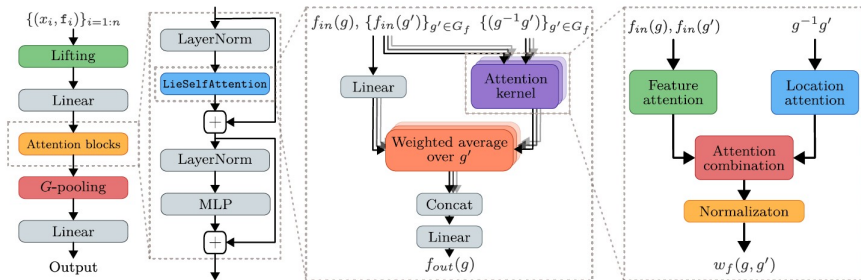


Figure 1. Architecture of the LieTransformer.

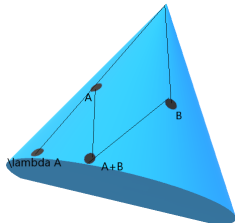
Source: [Hutchinson et al., 2021](#)

Geometry of the space $\text{SPD}(n)$

- ▶ Consider the vector space $S(n)$ of symmetric matrices
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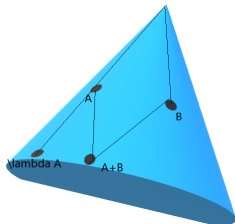
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 The set of all positive semi-definite matrices is a convex cone $\mathcal{C}$ in $S(n)$: for any $A, B \in S(n)$, $\lambda > 0$ one has $A + B \in S(n)$, $\lambda A \in S(n)$.



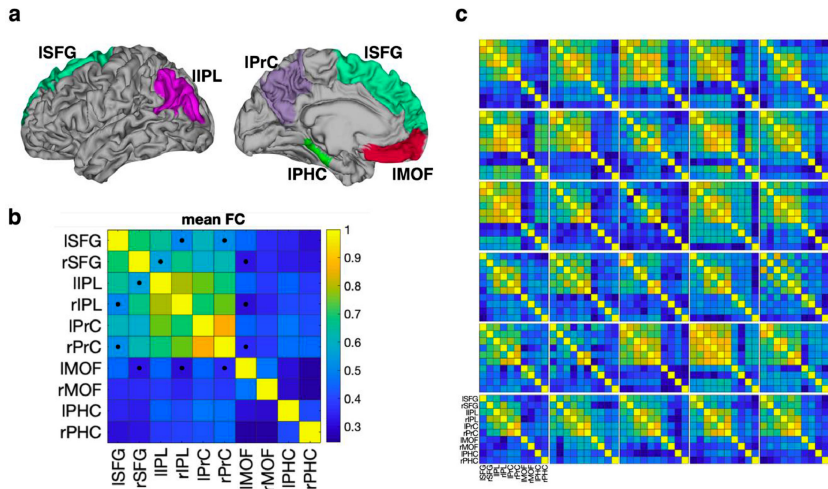
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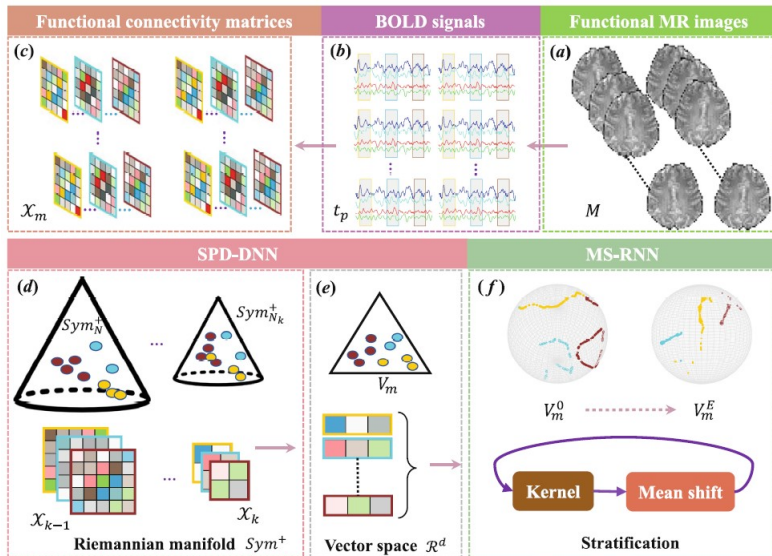
- ▶ Consider $\text{SPD}(n)$, the set of **symmetric, positive semi-definite and invertible** matrices (SPD=symmetric positive definite). This is an open set in the cone $\mathcal{C}$. Thus, $\text{SPD}(n)$ has a natural structure of manifold of dimension $\frac{1}{2}n(n+1)$ (e.g. [Moakher and Batchelor, 2006](#)). This manifold is actually [the interior of a pointed convex cone](#).

SPD-matrices: brain connectomes

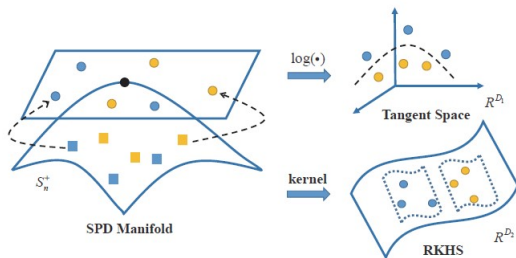


Source: You et al., 2021

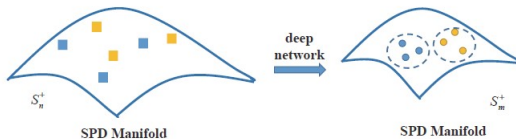
SPD-matrices: detection of brain state changes



SPD-matrices: application to face recognition



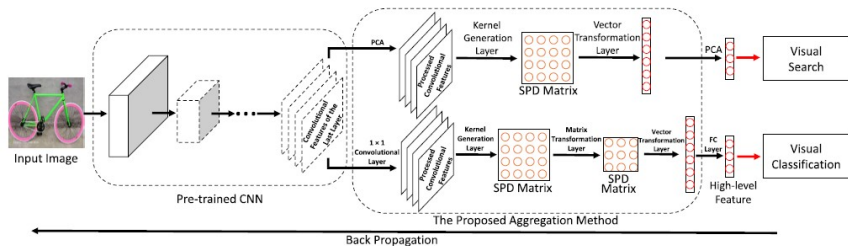
(a)



(b)

Source: [Dong et al., 2017](#)

SPD-matrices: CV tasks



Source [Gao et al., 2019](#), see also [Gao et al., 2017](#)

Other references of applications involving the manifold $\text{SPD}(n)$

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